

How have disruptions in global grain supply (due to the Ukraine conflict) influenced food price inflation and food security in Sub-Saharan Africa and South Asia?

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1. Introduction

The conflict between Ukraine and Russia, which has been borne out of the annexation of Crimea in 2014 and accelerated significantly in terms of fullscale invasion, has had a far reaching impact on the world food system. Although the issue of food security is always jeopardized during any such conflicts that occur in any region, the Russia-Ukraine war has crossed all regional boundaries, demonstrating how loosely dependent supply chains and agricultural trading is globally. The worsening of the conflict interfered with the production and export of several major agricultural products, significantly changed the world food prices, and exacerbated the existing weaknesses in the food systems, particularly in low-income and import-reliant states (FAO, 2023).

Before the war, Russia and Ukraine accounted for almost 30 percent of world wheat exports, 20 percent of corn and more than 70 percent of sunflower oil (OECD-FAO, 2022). The area is also important in exporting the fertilizers. Since Russia is the largest exporter of nitrogen fertilizers and natural gas in the world, the country makes too big an impact on the production of food in the world indirectly through agricultural inputs (FAO, 2022). Ukraine, in its turn, being called the breadbasket of Europe, contributed significantly to feeding not only the population of Europe, but also of countries found in Africa, the Middle East, and Asia. The rise in the prices of the basic foodstuffs and political and social unrests in the vulnerable nations have hence been the cascading effects of disruption in the movement of grains and fertilizers in this region.

Ukraine is a notable producer and exporter of various essential agricultural commodities like energy resources, fertilizers, etc. It plays a major role in the global agriculture and food export markets including wheat, maize, barley, oats, etc. Ukraine also produces 30 percent of

the world's wheat, 20 percent of maize, and 70 percent of sunflower oil approximately. It also one of the 'top three global exporters' of wheat and maize.

Countries in South Asia and North Africa have been particularly vulnerable since the two depend heavily on Russia and Ukraine as markets to import grains, edible oils, and fertilizers. Price volatility in the said areas has been aggravated by pre-existing socio-economic vulnerabilities, such as climate shocks, debt, and income inequality. As an illustration, more than 80 percent of the wheat supplied in Egypt was carried out by Russia and Ukraine pre-conflict (IFPRI, 2022). Similarly, other countries on South Asia such as Bangladesh and Sri Lanka had not only experienced food inflation but also fuel and fertilizer deficiency which halted their agricultural output (World Bank, 2023).

A question which was highlighted by the war was the use of food as a weapon of politics and strategy. The temporary blockade of the Ukrainian ports by Russia, including the major port of the Black Sea, Odessa, and delays and uncertainty on the Black Sea Grain Initiative disclosed the fragility of food supply chains worldwide to conflict-related manipulation (CSIS, 2023).

In this paper we get into the war effect within the food system of the world by discussing the breach in the production, trade, prices and accessibility of food. It also examines the particular impacts on food security in South Asia and North Africa, and it gives an evidence-based opinion on how the geopolitical dispute gave rise to a humanitarian food crisis and also expanded world food disparities.

Choosing this topic came from a strong interest in global economics and how international relations and events can impact everyday lives around the world. The Russia-

Ukraine war caught my attention not just as a geopolitical conflict but also because of how it has disrupted food supplies, in particular grain like wheat, maize, etc. which many countries rely on. I was also curious to know how this conflict between Russia and Ukraine has effected people in places like Sub-Saharan Africa and South Asia where food prices shot up making life tough for many people. Researching on this topic has allowed me to correlate my interest in global affairs with real world economic issues. It also helps me in understanding how closely the world is connected in terms of something as basic as food.

2. Shocks to Production & Supply

The current part analyses how the war between Russia and Ukraine has derailed agricultural outputs, supply, and logistics of nourishment as well as input materials, mainly fertilisers, which have caused ripple effects on global food security.

2.1 The Agricultural Sector of Ukraine is being Attacked

There has been the greatest disruption in the agriculture industry of Ukraine. By July 2025, the war made nearly a quarter of Ukrainian agricultural land either unavailable (due to occupation or unsafe to use due to unexploded ordnance) or inaccessible, amounting to some 138,000 km² (about 22 million acres) of potential farmland unavailable. The crisis was further aggravated by lack of labor: it has been estimated that one third of agricultural labour force had been conscripted or displaced . This has further resulted in losses of production through destruction of infrastructure such as grain elevators and machinery, especially in the east (theguardian.com).

The most dramatic illustration is the June 2023 destruction of the Kakhovka Dam that eliminated irrigation of 584,000 ha that will result in a loss of 4 Mt (grains and oilseeds) or 4

percent of Ukraine grain production annually. Total grain production in Ukraine was produced at 86.7Mt in 2021/22 and 61.8Mt in 2022/23, representing a sharp 29 percent decrease, despite the dam collapse ([consilium.europa.eu](https://consilium.europa.eu/en/press/press-releases/2022/05/20220510-ukraine-grain/)).

The global markets are directly impacted by such losses; Ukraine had been the fifth-largest exporter of wheat in the world, the important supplier of maize and sunflower oil . Less seeding 22 percent of spring 2022 hectares will lead to no doubt to the persistence of output shocks even after the active conflict periods ([consilium.europa.eu](https://consilium.europa.eu/en/press/press-releases/2022/05/20220510-ukraine-grain/)). Moreover, the seizure of agricultural inputs and equipment and stealing of grain (more than 400 t) in occupied territories have added to the loss that is valued at US \$4.3 b in equipment and crop damages.

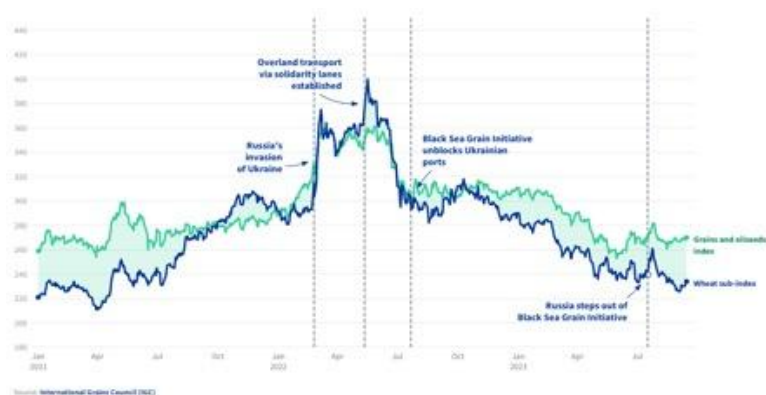
2.2 Disruptions to Ukrainian Export Infrastructure

Prior to the war, Black Sea ports managed more than 90 percent of Ukrainian grain exports ([sciencedirect.com](https://www.sciencedirect.com/science/article/pii/S0924646022000000)). The blocking of ports by the Russian navy, including Odesa, Chornomorsk, and Pivdennyi, blocked commercial shipping exports in the early 2022, and its volume decreased by more than 90 percent in March through May 2022 compared to 2021 ([consilium.europa.eu](https://consilium.europa.eu/en/press/press-releases/2022/05/20220510-ukraine-grain/)). Consequently, the volume of exports has dropped to ~2 Mt/month, previously, in the times before the war, ~7 8 Mt/month (apps.fas.usda.gov).

The Black Sea Grain Initiative (BSGI) which has continued to work between July 2022 and July 2023 and allowed the release of almost 33 Mt of grain and foodstuffs, mostly maize (51 percent) and wheat (27 percent) ([consilium.europa.eu](https://consilium.europa.eu/en/press/press-releases/2022/05/20220510-ukraine-grain/)). Although it brought back relief in the world, Russia exiting caused disruptions once again to be faced by Ukraine with an alternative redirected by road through EU solidarity lines and Danube ports

(consilium.europa.eu). By 2023, maritime share was restored to ~80 percent, but costs and logistic complexity were on a high level .

Aggregate export prices have also been reducing rapidly: the total grain export has fallen to 49 Mt in 2022/23, compared to a five-year average of 47.2 Mt, and is expected to be 32 Mt in 2023/24, a 32 percent reduction. The externalities in the supply chain were a major reason of tightness and global cereal market volatility in prices.



IMECiD – Incubator for Media Education and Development. (2023). *Annual evolution of the FAO Food Price Index by group of products, from 1990 onwards [Graph]. Based on FAO data.* <https://imedd.org/ukraine/global-food-security-ukraine-war/>

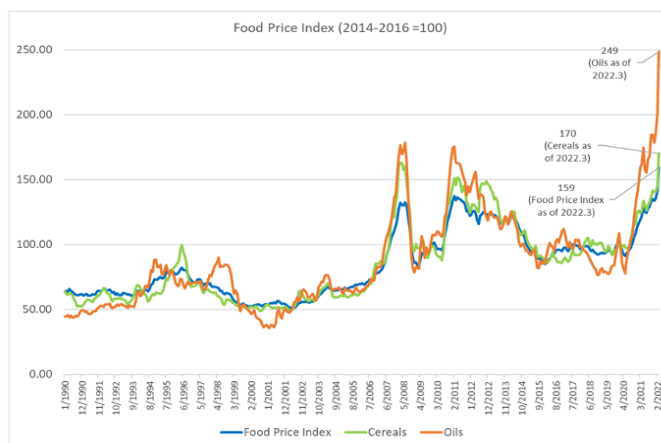
This infographic illustrates the deformation and re-organization of the grain export chains of Ukraine between 2021 and 2023. It graphically explains how there is a drop in the Henry-Loptson index of Black Sea maritime exports at the time when Russian naval blockades happened and how consequently, the alternative export channels through land and Danube river entrances pave the way due to the EU Solidarity Lanes initiative. It gives the much needed background on how to interpret the logistical bottlenecks on the available global grain supply.

2.3 Global Cereal Price Spikes & Disruptions

2.3.1 Price Index Surges

Cereal and food price indices increased at a global scale after the invasion. The FAO Food Price Index increased during early 2022 to reach a high of 159.3 at the end of March 2022 (12.6 percent YoY) and even higher in the world (jircas.go.jp). Cereal Price Index increased by a significant scale of 17.1 percent in March and this was the highest level since the year 1990 (jircas.go.jp).

The rise in prices was sharp: according to IGC data, the price of wheat in March 2022 rose by 58 per cent and the broad cereals price by 34 per cent compared to a year ago, in March 2021. Price pressures were also reinforced by energy prices and fertilizers shortage; fertilizers have not been available at the same level since mid-2022 due to the sanctions and the logistics issues.



Food and Agriculture Organization of the United Nations. (2023). FAO Cereal Price Index.

FAO World Food Situation. <https://www.fao.org/worldfoodsituation/foodpricesindex/en/>

This is a trend graph indicating the world cereal prices growth or decline on monthly basis since 2019 and by 2023. It points out the dramatic rise in the prices of cereal products (especially wheat and maize) due to the Russian invasion of Ukraine in February this year. This index had reached its highest point in March of 2022 and was volatile until the year 2023 due to mean interrupted supply chains and constricted communications concerning the possibility of exports being conducted in the Black Sea region.

2.3.2 Examples of Indices

- FAO Food Price Index 2019-2023 (Figure 2.3A): There has been historical peak in February 2022 .
- FAO Cereal Price Index 2020-2023 (Figure 2.3B): shows a March 2022 spike by more than 160, and a continued rise into 2023.

2.3.3 Ukrainian Transmission to Beyond Ukraine

Seismic waves travelled around the world. The fact that Russia is one of the largest suppliers of fertilizers in the world, combined with sanctions and difficulties with logistical capacity, led to a noticeable decline in supply and a significant worsening of price inflation in all areas of agriculture. High costs of fertilizer raise the cost of production directly: energy and fertilizer costs make up to 50 per cent of the variable cost of cropping, in the highly developed ag economies. This meant that the farmers in the US, EU, India, and other countries would either use less input or would be late to plant which impacted the yields and food supply. Export control in many importing states further destabilized the markets, and detoured to inflationary pressures with varying degrees of success in balancing the prices in a country (apnews.com).

2.3.4 Persistent Volatility

There was also still volatility, even though it dropped in the short term following the opening of Summer 2022 corridors. When BSGI collapsed in the autumn of 2023, the uncertainty returned; world grain prices went up once more (apps.fas.usda.gov). Add to this that although indices fell slightly, they were substantially higher than before 2020, still more than 24 percent higher than 2019 averages (wandilesihlobo.com).

3. The Impact of Food-Security: Ukraine and the World

Section 2 threw light on the dramatic supply shocks due to Russian-Ukraine war. In this part, we consider how those shocks have been converted into food-security impacts, both in Ukraine itself and internationally, through the main aspects of food security: availability, access, utilization and stability.

3.1 Ukraine Domestic Food Insecurity

3.1.1 The Scale of Need and Physical Access

In Ukraine, the disruption to agricultural operations, damage to infrastructure facilities, and forced displacement of population have created a severe food situation in the country. There is currently food insecurity among up to one in every three Ukrainians, especially in the conflict-affected locations, because of insecure access to markets, destroyed supply chains and closed stores (wfp.org). In the World Food Programme, about one-third of the inhabitants around frontlines are experiencing challenges accessing food because of destruction, service interruptions, or closure of foods markets . Such problems are further aggravated to those that are displaced internally, as most of them stay in makeshift shelters and do not have regular supply of food.

Food shopping has become much challenging: as a survey of ~25 percent of Ukrainian households interviewed in mid 2023, citizens had been struggling to keep higher food prices at one point in their lives (reuters.com). Due to the expensive food prices that have been observed locally- coupled with the higher costs of fuel and other sources of input, the state of the families has had to reduce on its meals, or reduce their amount, or go into a debt condition in order to sustain their food costs (wfp.org). According to the official poverty levels, Ukraine is currently about 29 percent with a very serious socio-economic decay (reuters.com).

3.1.2 Humanitarian Response & Market Functionality

The crisis has not stopped markets in most of Ukraine but regulating access when food can be obtained locally. World Food Programme has also introduced more than US\$ 1.3 billion since March 2022 through local purchase and cash aid that has facilitated the mobilizing of Ukrainian producers and benefitting producers and consumers ([wfp.org](https://www.wfp.org)).

However, humanitarian missions are highly risky: last year, more than 40 assaults were carried out on the WFP checkpoints or distribution channels, exacerbating the problem of delivery (wfp.org). Failed to stabilize the turbulent situation along the frontline, the latter leads to poor continuity in food accessibility at least in the east and south where concrete fighting had ruined the roads and food infrastructure. On the whole, Ukraine is plagued by two-fold danger: there are no chances that the ethical perspectives of providing food couldn't be overcome as society lacks food sources and financial capabilities to purchase them because of the destroyed economy, mass displacement, and ruined infrastructure.

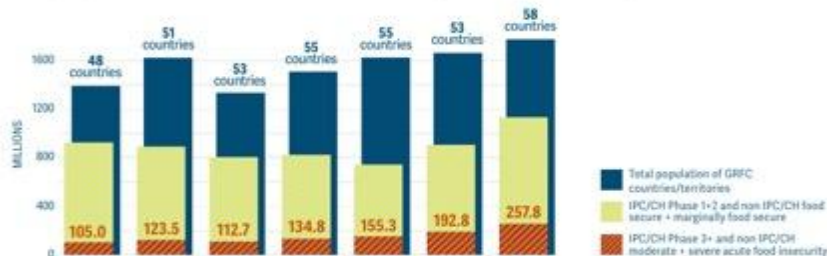
3.2 Global Acute Food Insecurity

3.2.1 Rising Numbers & Breadbasket Losses

In early 2022, there were about 135 million people across the world with acute food insecurity before the Russian invasion. This number had increased to approximately 276 million, or more than two-fold the foreign debt in the middle of 2022, which is the highest it is ever reported (un.org). It has been estimated the total amount of individuals affected by price and supply shocks in the conflict on the planet was between 1 and 1.7 billion; this is moderate to severe food insecurity ([sciencedirect.com](https://www.sciencedirect.com)).

This sharp increase is directly connected with the unexpected interruption of wheat and maize exports of the amalgamated Russia-Ukraine breadbasket directly affecting nearly 30 percent of international wheat and barley, 20 percent maize, and more than 50 percent of sunflower oil exports (concernusa.org). Inflation, high access, and less nutritious food alternatives were contributed by cost acceleration of commodities.

Number of people in GRFC countries/territories facing acute food insecurity, 2016–2022



Food Security Information Network. (2023). Global report on food crises 2023. World Food Programme. <https://www.fsinplatform.org/global-report-food-crises-2023>

This graph visualizes how many people are at risk of acute food insecurity (IPC/CH Phase 3+ or worse) by the year in GRFC countries and territories since the year 2016 to

2022. It is illustrated in the data that there was a massive increase in the number of people experiencing food insecurity especially post-2020, and that in the year 2020, the number of people with food insecurity stood at 155.3 million which soared to 257.8 million in the year 2022 (2-years post-Russia Ukraine war outbreak).

The deterioration of the situation is mostly explained by the mix of conflict, economic shocks (and in particular by price inflation driven by war) as well as by extreme weather events. The graph also separates the total population, food-secure or marginally food-secure group (yellow), and the acutely food-insecure group (red) and highlights the increasing gulf between world food security.

3.2.2 Humanitarian Stretch Across the World

Humanitarian organizations face pressure like never before. As in the case of the World Food Programme, whereby the operational prices have risen by US\$73.6 million per month- an increment of 44 percent in comparison to the 2019 averages- this increase in price is largely associated with food and fuel prices (csis.org). The International Rescue Committee estimated that in 2022 alone 47 million other people would be exposed to acute hunger owing entirely to the Ukraine war, populations in Yemen, the Sahel, the Horn of Africa, and Afghanistan (rescue.org).

These shocks are particularly disastrous in places that are already stressed:

- ***Horn of Africa:*** in Somalia and Ethiopia, where the drought was severe, there were acute food crises. In Somalia, almost 90 percent of its wheat needs are provided by Ukraine/Russia, whereby fluctuations in wheat prices instantly turned into famine-like situations (rescue.org).

- **North Africa/Middle East:** The countries that import over 50 percent of wheat, e.g. Yemen, Sudan, etc. experienced inflation of bread prices with the result of social unrest (concernusa.org).

3.2.3 Price Shocks Linkage and Coping Strategies

It has been demonstrated that a 1 percent increment in world food prices has the capacity to drive 10 million more to the state of abject poverty (mdpi.com). Since the food price indices increased by 30 to 60 percent in the immediate aftermath of the war, this meant 100+ million people slipping into poverty besides constant humanitarian disaster situations. In early 2022, the FAO Food Price Index reached the highest rates it ever did, and to the end of 2023, it is still 12-13% higher than the prewar levels (economicsobservatory.com).

The coping measures embraced by countries included export controls, food prices, and emergency in-kind cash outflow. These measures provided temporary relief and in most cases increased speculation in the world markets. Stock- and sovereign reserve holders (e.g. China, which bought up to 50 percent world wheat supply in early 2022) mitigated the impacts of rapid price increases on the local supply of certain foods--but increased price competition on the global market and made low-income countries more dependent.

3.3 Socioeconomic and Gendered Consequences of Food Insecurity

Populations have not experienced the effects of food insecurity caused by the Russia Ukraine war in the same way. Poor resilience and confirmation of pre-existing inequality expose vulnerable people to disproportionate risks such as women, children, small holder farmers, and urban poor. This subsection puts into focus how gender and socioeconomic status influences the burden of the food crisis as it occurs in Ukraine and in the whole world.

3.3.1 Women and Girls: The Early and the Last Harvest

Women living in food-insecure situations are by far the shock absorbers in households, particularly the low-income. Statistics demonstrate that women globally face food insecurity at an increased level compared to men 27.6 percent and 25.4 percent of women and men, respectively, reported in 2022 by FAO (FAO, 2023).

Households that are headed by displaced women were the most food insecure in Ukraine, this was because they had lower access to income, lost jobs and uncompensated care during displacement. Most of the women have said that they ate less in order to feed more children which is also observed in terms of COVID-19 (WFP, 2023). Such prioritization in nutrition has unseen health outcomes consisting of anemia, under nutrition, and reproductive health.

In Sub-Saharan Africa and South Asia where women tend to be the ones who provide food to their families inflation meant that fewer people ate, children became malnourished and girls in particular would be placed out of school as their families chose to buy food at the expense of their education.

3.3.2 Urban Poor and Informal Workers: Much exposure

The poor in urban areas in countries such as Lebanon, Egypt, and Bangladesh did not fare well in the face of the bullishness in prices of wheat and oil. The urban poor households, whose livelihoods were usually depending on daily wages, were hardly able to cope in case of the staple food prices increase by 30 to 50 percent in the first half of 2022 (UNDP, 2022). Households in informal activities who had no safety net found it hard to afford subsidized foods.

The role of the World Bank World Bank The World Bank is a development bank founded in 1944 and based in Washington. It is a member of the World Bank Group. Hermes Replica belt The poorest fifth of the population in Bangladesh ate almost 60 percent of all their money on food at the mid-point of 2023. This is backward and is very risky in that, it forms downwards illnesses in terms of debts, malnutrition and susceptibility to health emergencies.

3.3.3 Malnutrition risk in children

Increased food insecurity has adverse effects on child health, especially in the already crisis prone areas. The headcount of children with severe wasting across the globe climbed to a high of more than 45 million in 2022, according to UNICEF (2023) which indicated that future prospects will unfavorably rise since use of fertilizers is declining due to insufficient fertilizer supplies hence decreasing crop productivity.

Conflict displaced children living in Ukraine are exposed to more than food deficiencies; they are also at long-term risks of health and structural problems. In East Africa, the decline in the availability of wheat was largely profound and this promoted the use of replacement foods, with high-calorie density, low nutrients, which led to stunting and undernutrition among children younger than five years old.

3.3.4 Blind Spots on Policy and Gaps in Social Protection

Export ban, fertilizer subsidies, and food rationing were most of the responses of the government but proved macro plans, and did not touch on intersectional vulnerabilities. Minor nations increased food benefits, school foodstuffs, or gender-sensitive aid initiatives during the high points of the snag (UNDP, 2023).

Unless there are inclusive protection measures, the issue of food insecurity will widen the current inequalities. According to one of the reports by the UNDP:

Access to calories is only a part of the food crisis: and that is inequality of access to resilience.

4.1 Cost of energy and fertilizer input costs

The war which has caused a high rise in the prices of energy and fertilizer- vital components in crop production across the globe- did so even before the grain shortages. Gas prices rose by an estimated 250% in Europe which is a gas importer that relies on Russian gas between early 2021 and mid 2022 putting more pressure on nitrogen-based fertilizers, which are linked to gas prices (IEA, 2022). This domino effect caused a diminishing source of fertilizers and it had a direct implication of the cost of farming in all continents.

Russia is a significant exporter of nitrogen and potash fertilizers, interruption and embargo on trade raised the price of urea, DAP, and potash by 70-100 percent as of mid-2022 relative to that of the previous year (FAO, 2022). U.S., EU, and Indian farmers claimed a tremendous decline in the use of nitrogen with some using 20 or 25 lower amounts of nitrogen-based fertilizer, which sprung directly to endanger yields (IFPRI, 2023).

In India and Bangladesh, where governments compensate market prices with fertilizer subsidies, the inflation was first transacted by governments. Nonetheless, such costs were later passed to farmers, and the prices of the privately imported goods have remained as high as 30-50 percent due to the costs, causing farmers to plant less or experience low harvests, particularly of rice and wheat (World Bank, 2022).

There are the two ripple effects of this input-cost shock:

- ***Short-run:*** Adequate cost due to increased prices makes farmers postpone seeding or cut on the use of fertilizers, which restricts the production of crops.
- ***lagged:*** Reduced yields trigger market unpredictability in the ensuing harvest season, yet again raising food insecurity and price instability.

Overall, the price shocks in energy and fertilizer were really a multiplier as they intensified the effects of the blockades in ports and the destruction of the land (under Section 2) and they also squeezed farmers and consumers around the world.

4.2 Export prohibitions and bans & trade restrictions

After the hike in international prices, there were the restrictive trade policies that were implemented by different countries which enhanced the constraint of supply and volatility even more.

There were export controls on:

- In May 2022, India temporarily halted wheat export.
- In June 2022, Argentina added export quotas on maize and wheat.
- In the harvest of 2022, Kazakhstan restricted the exports of wheat by 30 percent.

These measures only protected consumers within the country but increased world price spikes, particularly in areas, which depended highly on imports of these exporters (FAO, 2022).

The temporary (introduced in April 2022 but continuing into 2023) restrictions on fertilizers and wheat imposed by Russia meant that it significantly reduced supply. The panic

reaction played out on the market through speculation even though some of the measures were lifted.

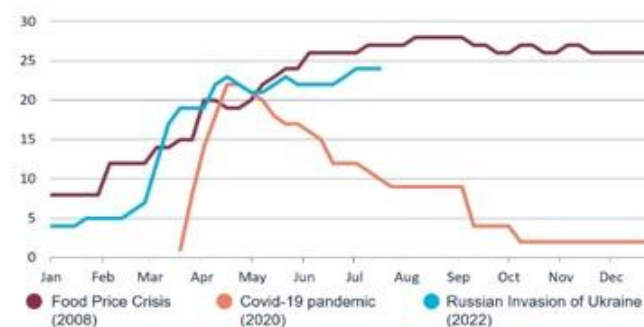
The export restrictions in Ukraine such as port closure and slow export licenses proceeded to halt or postpone the exports of grains, oils, and seeds (Council EU, 2023). Once the alleviation of the blockades with the help of the Black Sea Grain Initiative passed, the end of the scheme in July 2023 provoked new blockages in exports and uncertainty in the market.

There were systemic impacts of these dynamic trade restrictions:

- Price volatility: abrupt changes in price as markets would sky rocket when a ban was called and partly fall when it was cancelled.
- Loss of confidence in the system of international trade-buyers wanted to have several suppliers leading to speculative hoarding.
- Others such as Bangladesh who depended on Eastern Europe to import maize/oil seeds had to turn to alternatives which may prove very expensive or their local substitutes.

This political context led to the inability to localize the consequences of the war to only Ukraine because of its effect on the availability and affordability of food all over the world.

Figure 1: Number of countries with active food export restrictions



Source: IFPRI Food and Fertilizer Export Restrictions Tracker
Retrieved July 2022

Department of Agriculture, Fisheries and Forestry (Australia). (2022, August 10). Food export restrictions rise in 2022 [Chart]. High food prices drive export restrictions.
https://www.agriculture.gov.au/about/news/high-food-prices-drive-export-restrictions-aug-22?utm_source=chatgpt.com

This graph is taken by the department of agriculture of Australia, and it shows how many countries imposed the export restraint during the three crisis categories food price crisis 2008, the COVID 19 Pandemic, and the Russia Ukraine War (2022). It indicates that there is a massive rise in export ban in 2022, 32 countries, 26 of which remain ongoing, in reaction to inflation of food prices due to war.

4.3 Overlap of Pandemic

The effect of the war overlapped with remnants of weaknesses due to the COVID 19 pandemic, clearly perfecting the storm of food systems.

In 2020-21, there were lockdowns during the pandemic, logistics were disrupted as machinery and factories were closed whereas the shortage of labor force occurred. The smallholder farmers were lagging behind in picking up as the demand and wages declined especially in India and Southeast Asia (UNCTAD, 2022). Before the Ukraine war hit, gaps in manufacturing, transport, and logistics were aggravated by the lack of stability in the supply chains.

Both COVID testing and the war problem caused additional border delays. For instance:

- China road and India disruptions China India road and mid 2021 affected transit route to India fertilizer.

- Ukraine routes via Poland and Romania land routes were not able to handle the increased volume after the blockade of Black seas.

All these shocks did not permit a quick recovery of supply chains. As ports began to reopen, rates were also hampering the effective transport of grains with costs averaging at four and a half thousand dollars per container in late 2022 compared to just one and eight pre-pandemic (UNCTAD, 2023). Additionally, on tight government expenditures due to pandemic aid, rescue programs of small farmers or stabilize prices failed or were underfunded to enhance societal resistance to food crisis due to the war.

4.4 Global Food Systems Structural Weaknesses of Food

Intertwined war, energy shock and the effects of the pandemic left structural weaknesses in the global food system. Such weaknesses justify why local shocks in Ukraine snowballed into general insecurity on different continents.

4.4.1 Monopoly of some exporters

This is a weakness because the country is being overly dependent on a very limited number of exporters. Only a few suppliers are used in the global grain markets. By 2021, five nations were exporting more than 70 percent of the world wheat supply (FAOSTAT, 2022). The two countries produced close to 30 percent of world wheat and 70 percent for sunflower oil. This concentration in the market allowed that when something broke it caused supply panic and price spikes in the world market as those was experienced when the Black Sea ports were blocked.

4.4.2 Rigid Trade and Transportation systems

This is because the world food supply chain is mostly not resilient but cost-optimized. In the case of war closing the sea routes in the Black Sea, there were not many cushions available (a rail route through Poland or a road through Romania). Those mostly affected were the developing nations of Africa and Asia, countries that rely on mass imports of cereals and fertilizer since they had little local storage or varied sources of suppliers.

4.4.3 Weak Social Protection System

Food price or ration distributions or cash transfers in most of the low- and middle-income countries are not indexed to international inflation. Consequently, rising prices of wheat or cooking oils could not be followed in the domestic safety nets. Meanwhile, the growth of inflation in Bangladesh exceeded the rise of wages whereas in Egypt food subsidy programs were not able to take up the extra price of imported wheat without using foreign exchange volumes.

5. Russia-Ukraine War effects on the food security in the countries of South Asia

The war between Russia and Ukraine, which intensified in February 2022, resulted in significant ripple effects regarding the world food systems. The South Asia region whose chronic food security and macroeconomic vulnerabilities were previously incurred were disproportionately affected. High import dependencies coupled with weak agricultural infrastructure and shocks easily transmitted to people living in the countries like India, Bangladesh, Pakistan, Nepal and Sri Lanka which were the worst states that were impacted.

5.1 Disruption of Global Grain and Fertilizer Supply Chains

Both Russia and Ukraine are key exporters of basic products in the world. The two combine to export about 30 percent of the world wheats and over 70 percent of sunflower oil (FAO, 2022). Russia also exports fertilizers in large amount in the world. At the instant when trade in these countries is blocked by the war, trade in South Asia is faced with direct results: shortage in principal imports, rising prices, writhing market indicators.

Even though India produces sufficient wheat to feed its people, the use of fertilizers is imported. The war caused the interruption of gas-based fertilizers and potash supplies originated in Russia and Belarus. The Indian Ministry of Chemicals and Fertilizers indicated that urea prices within the international market increased by almost 70 percent between January and May 2022. Although government of India internalised this shock by increasing funds towards fertilizer subsidies, this could not be sufficient to sustain at least the smaller economies like Nepal and Sri Lanka.

5.2 Inflationary Pressures and Volatility of food prices

International food prices increased and these increases were conveyed directly to domestic inflation in South Asia. The World Bank (2023) argues that some countries, including Sri Lanka and Pakistan, realized food inflation of above 10 percent most of 2022 and 2023. This inflation affected the poor disproportionately since the latter spend high fraction of their income on food.

Food inflation in late 2022 in Bangladesh reached above 10 percent due to increasing prices of edible oil, wheat and pulses. The Bangladesh Bureau of Statistics found that more

than 20 percent of the households had to cut on the meal size and more than 5 percent skipped the meals altogether during the period.

This was especially acute in the case of Sri Lanka. In witnessing an economic crisis and a low foreign exchange reserves, the country was under severe import restrictions. In the middle of 2022, the nation could not afford fuel or basic food imports. FAO (2023) estimated that more than 6 million citizens of Sri Lanka were food insecure after the consequences of the war.

5.3 Reduction of Agricultural Investment and domestic food production

Against the regional agricultural investment was also pushed down by the economic uncertainties and inflatory environment. The smallholder farms in Pakistan were also exposed to increased cost of inputs such as seeds, diesel and fertilizer. The Pakistan Bureau of Statistics recorded a reduction of almost 20 percent in the rate of fertilizer application between the years 2021 and 2023.

Use of lower input reduced yield, although low yields were recorded in wheat and maize. In India where the government was ensuring the protection of farmers by providing them with Minimum Support Prices (MSP) and subsidies, yields were still going down in places due to late planting and the decreased use of fertilizer. According to the Central Bureau of Statistics, the agriculture sector in Nepal shrank by 2.3 per cent in FY2022.

5.4 Impact on Import-Dependent Nations and Trade Deficits

Other countries that depend highly on food imports are Bangladesh, Nepal, and Sri Lanka to import wheat and edible oil. These nations had a ballooning import bill due to soaring prices and low supplies especially on the international front.

Using the data provided by Bangladesh Bank, wheat imports in Bangladesh have dropped by more than 35 percent in FY2022 (v/s FY2021). That caused a shortage in the country, raised the price of the wheat flour and replaced it with cheap cereals.

Sri Lanka, which was suffering from a foreign exchange crisis unlike ever before, experienced a drop in the bill on food imports not due to its enhanced self-sufficiency but because it could not afford them any more. This put millions of people into food insecurity and poverty.

The fact that Nepal is a land locked country made it to be even more disadvantaged. The logistics expenses were high, and the use of Indian ports was time-consuming, which resulted in expensive imports. Its impact was the price transmissions into the domestic markets.

5.5 Socioeconomic Consequences and Increased Food Insecurity

Socioeconomic consequences of the food crisis related to the war were more extensive. The Global Report on Food Crises (2023) states that in 2021, the issue of acute food insecurity affected 135 million people in South Asia, while, in 2023, the figure exceeded 160 million people. This is comprised of chronic malnutrition as well as the temporary crises caused by sharp rises in prices.

Inflation and funding shortages interrupted school feeding programs, maternal and child nutrition programs, and food ration. As an example, the PM-POSHAN (mid-day meal scheme) in India had an issue in the procurement who could not get supplies when food prices increased. In Pakistan, the Benazir Income Support Program (BISP) was extended in 2023, but the purchasing power of the mechanism was damaged by continuous inflation.

Also, other differences such as gender gaps were expanded. According to World Food Programme (2023) women in food insecure homes had higher chance of skipping meals and declining dietary diversity to leave children and male individuals to feed first. It is not only affecting the present nutrition outcomes but also has the intergenerational consequence.

5.6 Policies and Government Actions

The governments of South Asia had a combination of fiscal, trade, and social protection measures as response. India made large food and fertilizer subsidizes. The prices were stabilized and accessibility was provided with the help of the public distribution system (PDS). The Pradhan Mantri Garib Kalyan Anna Yojana (PMGKAY) was advanced into most of 2022 with an aim of offering grains free of charge to vulnerable groups.

Bangladesh also lowered import tariffs on food products and it promoted bilateral trade agreements. These attempts were however brought down by the speculative tendencies on the market and poor storage systems.

IMF and World Bank provided their helping hand to Sri Lanka along with the Indian and Chinese bilateral support. There was emergency food distribution in towns and cities. Nepal has also extended food subsidy scheme to mountain and remote areas.

Although, these steps helped to avert the crisis, they revealed more fundamental weaknesses as over-dependence on imports, inadequate domestic production and weak safety nets of food systems.

6. Conflict in Ukrainian and the Disturbing Waves: Food Systemic and Development Implications in North Africa

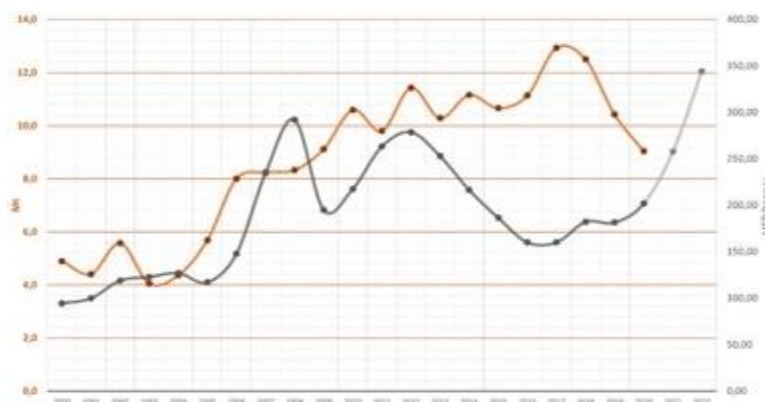
6.1 Strategic reliance on Russian and Ukrainian Imports

The Middle east and North Africa region-especially North Africa- are facing difficulties as the traditional sources of food such as land and water are limited and climatic practices cannot sustain food. States, like Egypt, Tunisia and Morocco, receive huge imports of wheat and other food goods, which is their major part in Ukraine and Russia. These two nations have provided more than 60 percent of the wheat imports in Egypt in 2021 (FAO, 2022).

This structural dependence was revealed by the war disruption of supply. Wheat exports were diverted, or cut off short, or delayed, whereby the result was local price accelerations. Governments of North Africa that provide food subsidies in order to achieve a social stability could not preserve the programs without burdening their budgets.

6.2 Sugar and wheat prices inflation in domestic consumption

The war and its tertiary effects on both energy and fertilizers increased global food prices to a critical level. Food inflation in North Africa was skyrocketing with food inflation in Egypt jumping above 30 per year in 2023, in Tunisia and Morocco rising above 15 per year (World Bank, 2023). This imposed an extra burden to countries that were already experiencing a slow economy and increasing government debt. Maintenance of food subsidies was made more expensive. Governments in at least a few instances decreased the amount of or coverage of these subsidies, building up a political backlash and unrest.



Wheat import volumes (Mt) and prices (USD/tonne) in North Africa from 2000–2022.

The chart illustrates rising dependency on wheat imports and the surge in prices after global disruptions—especially the 2022 Russia–Ukraine war.

Source: FAO (2023).

This figure presents the line graph with two variables, namely the volumes of wheat imports in North Africa measured at million tonnes, Mt on the left Y-axis, and average price of wheat imports against USD/tonnes on the right Y-axis in the period between 2000 to 2022. It points to the fact that the volume of imports was gradually rising until 2020, except for their acute volatility on prices, which was in 2022—after the war in Russia and Ukraine.

This is an indication of the fact that North African nations became more reliant on their wheat imports at a time when the prices were tumbling up and down in wild swings because of global shocks which included the 2008 financial crisis, 2011 Arab Spring and most recent of all the Russia-Ukraine war.

6.3 Problem of Import Financing and Currency Depreciation

When the foreign exchange reserves became low, and currencies were weak, the countries in North Africa experienced problems to finance their needs for importation.

Egyptian pound has depreciated almost by 50 percent in the last 2 years, which has influenced the increase in the prices of imports priced in dollars (IMF, 2023). This formed an ugly circle of an increased import bill and deteriorating inflation.

Egypt was the largest wheat importer in the whole world and the price of bread increased drastically even after the government tried to control the prices. Though far reaching, the subsidy system of the country was failing to match the high import prices.

6.4 Supply chain disruptions in Agricultural Inputs

Besides interrupting food exports, the war also did the same to the provision of the important agricultural inputs and fertilisers and fuel. The already fragile, water-scarce, and low yield North Africa agriculture further became susceptible.

Energy market volatility disrupted the production in the region as one of the importers of fertilizers, Morocco, is an exporter of fertilizers. The farmers in the area pushed back or gave up planting seasons as the cost of inputs was too high so the output at home fell.

6.5 Food Insecurity and Social Strains

The region was hit by food insecurity which escalated rapidly as the number of food-insecure individuals also grew as well as the severity of their cases. The Global Network Against Food Crises (2023) states that in the MENA region, almost 58 million people faced acute food insecurity in 2022, which is much higher than in the previous years.

Especially the urban populations who relied on market purchases were affected. There were bread riots and protests taking place in some areas, just like in the protest in the

2007/2008 food crisis. Emergency relief and short-term levels of exports bans were introduced to control domestic supply by governments.

6.6 Multilateral Aids and Policy Responses

North African governments undertook a number of mechanisms to counteract the crisis. These were increasing the social protection system, lowering the costs on foreign food, and bargaining new sources of importation (e.g. India, Brazil).

Multilateral organizations like IMF and World Bank emerged with financial aid. To stabilize their economies and viability of their food security infrastructures, countries like Egypt, in 2023 obtained a 3 billion IMF package to shore up their economies.

Nevertheless, there still exist long term weaknesses. Structural reforms are necessary to avoid over-reliance on external sources of supply, climate vulnerability, demographic pressures, etc. Sustainable agriculture--making solid investments in sustainable agriculture; Water-efficient irrigation systems--development of water efficient irrigation systems; Trade cooperation in the region--creating cooperation on the trade in the region.

7. Suggestions on Reducing Food Security Shocks in a conflict-driven international world

The war between Russia and Ukraine has highlighted the weakness of world food systems to geopolitical fluctuations. The ripple effects have been immense and especially on the import-dependent and low-income economies, in form of destruction of exports to inflationary pressures as well as less access to agricultural inputs. There is a need to act both

nationally, regionally and internationally to achieve resilience and long-term food security living in the conflict-driven world. This part provides constructive suggestions in six dimensions; trade, production, diversification, social protection, international cooperation and data systems.

7.1 The Reinforcement of Global and Regional Trade System

- ***Enforce Open, transparent trade policy:*** Exporters need to avoid the temptation to now and then institute export bans and instead operate within multilateral institutions such as the WTO to enact predictability. Open trade policies create confidence and ensure there is no panic buying and hoarding.
- ***Set up Regional Buffer Stocks:*** In regions like South Asia (through SAARC) or Africa Union these regional alliances should work together creating emergency grain stocks. The common buffer inventories may help level the market and provide prompt procedures during the time of emergency.
- ***Diversify trading partners:*** Countries who depend on imports should not be over dependent on a few exporters. Bilateral and multilateral agreements on trade ought to be pursued so as to gain access to a wider market in terms of suppliers, in Latin America, Southeast Asia and Africa.

7.2 Increased local agricultural and the security of input

- ***Invest in the Agriculture Sector:*** Nations must make more investments on irrigation, storage, transport and cold chain infrastructure. Modern solutions and post-harvest losses could be achieved with the aid of the public-private partnership.

- ***Secure Supplies of Inputs:*** An example is to maintain fertilizers and seed supplies, governments must contract long-term production tonnes, tap multiple source countries and develop domestic supplies. Such as, the expansion of urea plants in India and the new seed banks of Bangladesh are replicable.
- ***Invest in Climate-Adaptive Crops:*** High-yield varieties that perform well in an environment with changing climate should be favoured in agricultural research and development processes. Drought-tolerant wheat, short-term rice, and nitrogen-fixing legumes that thrive in soils that have degraded should reduce yield fluctuations caused by weather and conflict variations.

7.3 Increasing the Diversification of Food Systems

- ***Foster Dietary and Crop Diversity:*** Governments ought to encourage producers and consumers towards traditional and underutilized crops, including the growers of millet, lentils, and sorghum. This decreases the use of international traded commodities such as wheat and maize.
- ***Promote Urban and peri-Urban agriculture:*** Kitchen gardens, rooftop and vertical farming can make a difference in combating the lack of local food in urban areas and the overload of long-distance supply chains.
- ***Build Aquaculture and Livestock Industries:*** In developing sources of protein such as fisheries and livestock, one will be able to support food security especially in arid or land-constrained areas.

7.4 Enhancing the Social Protection and Safety Nets

- ***Scale up Food Assistance Programs:*** The countries are expected to escalate the food transfer programs, sheltered stay programs, school feeding, and the maternal nutrition programs during the crisis periods. Such programs as PMGKAY in India or BISP in Pakistan require inflation-based budget support.
- ***Implement Shock-Responsive Systems:*** More vulnerable populations can be cushioned by introducing dynamic social protection systems that can increase automatically when a crisis happens (i.e. Shock responsive, depending on a trigger e.g. hike in prices or change in weather).
- ***Digital Platforms:*** Digital technology, smartphone, digital identification and e-vouchers can make social transfers to be more efficient and transparent.

7.5 Enlarging International Cooperation and Multilateral Interaction

- ***Enhance Global Monitoring Systems:*** Mechanisms, such as the Global Agricultural Monitoring (GEOGLAM) initiative and AMIS (IGFS Agricultural Market Information System) should be enhanced in order to present real time information on the stocks of global food, global food prices and trade flows.
- ***International Finance Mobilize:*** The multilateral agencies such as the World Bank, IMF and IFAD ought to offer concessional loans and grants to the poor countries that import their food in order to continue necessary imports during emergencies.
- ***Conduct Coordinated Humanitarian Food:*** FAO and World Food Programme should liaise with local bodies to pre-place the aid in zones of vulnerability. The

smallholder farmers can also be supported through local procurement of the commodity of aid.

7.6 Data, Research and Early Warning Investments

- **Enhancing Food Security Data Infrastructure:** Society should invest in the data infrastructure by conducting household surveys on food and crop monitoring systems that detect early warnings, markets, and intelligence that identifies the changing risks, and thus governments are able to respond rapidly.
- **Facilitate Research Partnership:** Global collaboration in agricultural R&D could come up with area-specific answers to the issue in food security, such as output disease-resistant plants and effective irrigation methods.
- **Put in place Early Warning Systems:** The use of various information such as meteorological, market, and conflict-related data in food security early warning systems makes responses proactive. An example is the Famine Early Warning Systems Network (FEWS NET) which is a good model to be replicated.

Summary

The effects of the Russia-Ukraine war have exerted extensive implications in the biggest geopolitical arena but have essentially damaged the stability of the global food system. Being two of the largest exporters of wheat, maize, sunflower oil and fertilizers, the disruption of the exports of Russia and Ukraine resulted in a chain of supply shocks to the global environment. Such shocks together with previous susceptibilities compounded by the

COVID-19 pandemic and climate-related pressures constituted an ideal storm that has profoundly impacted food availability, affordability, access and usages in regions.

This study reviewed the multifaceted effects of the war on the food security in the global context and specifically presented the analysis of South Asia and North African regions that are especially exposed to the war effects because of their high vulnerabilities attributed by import dependence and the structural agricultural problems. South Asia had acute inflationary pressures, agricultural productivity was dwindled because of fertiliser unavailability and malnutrition levels were increased. In a similar manner, the North African nations with a high rate of dependence on AA (Russia and Ukraine) on wheat imports experienced skyrocketing levels of bread prices, weakening subsidy resources, forex crisis, and a significant increase in food-related social crises as a result of food inflation.

It was also analysed that the war enhanced food insecurity in tens of millions due to the importance placed on the issue by the low-income countries, notably the ones that are reliant on imported foods. The vulnerable populations plunged further into poverty and malnourishment as a result of price shocks. The physical supply of food was limited by bottlenecks in the supply chain, export control and unpredictable global food markets in conflicts zones as well as in import countries. Moreover, those were aggravated by the hoarding stimulated by the speculators, protectionist trade measures and fuel prices increasing, that undermined confidence in international trade systems.

Section 7 offered a suite of prospective policies, which was about calling to make long-term investments in agricultural research and development, diversification of regional trade and increased capability of domestic food production. It also stressed the need of emergency

foods, the need of increasing level of social protection, and developing multilateral cooperation to endure shocks in the food system in the future.

To sum up, the Russia-Ukraine conflict is a timely wake-up call to review how vulnerable the system of the global food architecture is. It is not only food security that requires a more equitable, diversified and resilient system, but also geopolitical stability and long-term sustainable development, founded on inclusive investment, proactive policy and global solidarity.

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